SAVEETHA SCHOOL OF ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA02-C PROGRAMMING

List of Programs to complete on 12.09.22

1. Write a program for matrix addition?

Sample Input:

Mat1 = 1 2

5 3

Mat2 = 2 3

4 1

Sample Output:

Mat Sum = 3 5

9 4

2. Write a program to find the number of composite numbers in an array of elements
   Sample Input;:
   Array of elements = {16, 18, 27, 16, 23, 21, 19}
   Sample Output:
   Number of Composite Numbers = 5
   Test cases:
   1.Array of elements = {26, 28, 37, 26, 33, 31, 29}
   2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
   3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
   4. Array of elements = {200, 180, 180, 270, 270, 270, 190, 200}
   5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100}

3. Write a program to print the number of vowels in the given statement?
   Sample Input:
   Saveetha School of Engineering
   Sample Output:
   Number o vowels = 12
   Test cases:
   1. India is my country
   2. All are my brothers and sisters
   3. Why dry sky
   4. Shy Try Cry
   5. EDUCATION

4. Write a program to print the number of negative numbers in an array of numbers
   Sample Input;
   Array of elements = {16, -18, 27, -16, 23, -21, 19}
   Sample Output:
   Number of negative numbers in Array elements = 3
   Test cases:
   1.Array of elements = {-26, 28, 37, -26, 33, -31, -29}
   2. Array of elements = {1.6, 1.8, 2.7, -1.6, 2.3, -2.1, .19}
   3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
   4. Array of elements = {-16, 2.8, -7, -1.5, 2.8, -2.8, -.19}
   5. Array of elements = {-160, -160, -180, -270, -160, -230, -210, 1-90, 0}

5. Write a program that would sort a list of names in alphabetical order Ascending or Descending, choice get from the user?
   Sample Input:
   Banana
   Carrot
   Radish
   Apple
   Jack
   Order(A/D) : A
   Sample Output:
   Apple
   Banana
   Carrot
   Jack
   Radish

6. Program to remove duplicates from the sorted array
   Sample Input:
   rray = {15, 14, 25, 14, 32, 14, 31}
   Sample Output:
   ed Array = {14, 15, 25, 31, 32}
   Test cases:
   1. {16, 16, 16 16, 16}
   2. {0, 0, 0, 0}
   3. {-12, -78, -35, -42}
   4. {1,2,3,7,8,9,4,5,6}
   5. {1-2,2-3,3-4,4-5,5-6}

7. Find the Mean, Median, Mode of the array of numbers?

   Sample Input:

   Array of elements = {16, 18, 27, 16, 23, 21, 19}

   Sample Output:

   Mean = 20
   Median = 19
   Mode = 16

   Test cases:

   1.Array of elements = {26, 28, 37, 26, 33, 31, 29}
   2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
   3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
   4. Array of elements = {200, 180, 180, 270, 160, 270, 270, 190, 200}
   5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100, 100}

8. Write a program to reverse an array

   Sample Input;:
   Array of elements = {16, 18, 27, 16, 23, 21, 19}
   Sample Output:
   Reverse Array elements = {19, 21 23, 16, 27, 18, 16}
   Test cases:
   1.Array of elements = {26, 28, 37, 26, 33, 31, 29}
   2. Array of elements = {1.6, 1.8, 2.7, 1.6, 2.3, 2.1, .19}
   3. Array of elements = {0, 160, 180, 270, 160, 230, 210, 190, 0}
   4. Array of elements = {200, 180, 180, 270, 270, 270, 190, 200}
   5. Array of elements = {100, 100, 100, 100, 100, 100, 100, 100}

9. Write a program to search the given element and display its position in a linear array.

   Sample Input;:
   Array of elements = {16, 18, 27, 16, 23, 21, 19}
   Element to search = 23

Sample Output:
Given element 23 is found at 5 th position

10.Write a program to search the given element using binary search method and display its position in a linear array.

   Sample Input;:
   Array of elements = {16, 18, 27, 16, 23, 21, 19}
   Element to search = 23

Sample Output:
Given element 23 is found at 5 th position